

## CSE563 — Assignment 3

### Unit III: Probability and Statistics

**Maps to CO3** (apply the concepts of probability and statistics for inference and regression analysis).  
**Marks: 50.** Attempt all questions. Standard normal table values are supplied where needed.

| Standard normal values supplied for this paper                                  |        |        |        |        |        |        |        |        |
|---------------------------------------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|
| $z$                                                                             | 0.5    | 1.0    | 1.28   | 1.5    | 1.645  | 1.96   | 2.0    | 2.58   |
| $\Phi(z)$                                                                       | 0.6915 | 0.8413 | 0.9000 | 0.9332 | 0.9500 | 0.9750 | 0.9772 | 0.9950 |
| $t_{0.025, 9} = 2.262, \quad t_{0.025, 15} = 2.131, \quad t_{0.025, 3} = 3.182$ |        |        |        |        |        |        |        |        |

### Part A — Short Answer and Definitions (10 marks, 2 each)

**Q1. [2 marks]** Distinguish between mutually exclusive and independent events. Can two events with non-zero probability be both?

**Solution.** **Mutually exclusive:**  $P(A \cap B) = 0$  — they cannot occur together. **Independent:**  $P(A \cap B) = P(A)P(B)$  — knowing one occurred does not change the probability of the other. **No.** If  $P(A) > 0$  and  $P(B) > 0$  and they are mutually exclusive, then  $P(A \cap B) = 0 \neq P(A)P(B) > 0$ , so they are dependent. In fact they are maximally dependent:  $P(A | B) = 0$ .

**Q2. [2 marks]** State the Central Limit Theorem and explain what “standard error” means. If a sample of 100 has  $\sigma = 20$ , what is the standard error? How large a sample would halve it?

**Solution.** **CLT:** for i.i.d. samples of size  $n$  from any distribution with mean  $\mu$  and finite variance  $\sigma^2$ , the sample mean  $\bar{X}$  is approximately  $N(\mu, \sigma^2/n)$  for large  $n$ . The **standard error**  $\sigma/\sqrt{n}$  is the standard deviation of the *sampling distribution* of  $\bar{X}$  — it measures how much  $\bar{x}$  would vary between repeated samples, not how much the data vary.  $SE = 20/\sqrt{100} = 2$ . To halve it to 1 we need  $\sqrt{n} = 20$ , i.e.  $n = 400$  — four times the data for twice the precision.

**Q3. [2 marks]** A 95% confidence interval for a mean is  $[12.4, 15.6]$ . State precisely what this means, and state one common misinterpretation.

**Solution.** **Correct:** the interval was produced by a procedure that, over repeated sampling from the same population, captures the true  $\mu$  in 95% of samples. The width also tells us  $\bar{x} = 14.0$  with margin of error 1.6. **Common misinterpretation:** “there is a 95% probability that  $\mu$  lies between 12.4 and 15.6.” In the frequentist framework  $\mu$  is a fixed constant: it is either in this particular interval or it is not. A probability statement about the parameter requires a Bayesian *credible* interval.

**Q4. [2 marks]** Define Type I and Type II errors. In a cancer-screening test, which is more serious, and how would you adjust  $\alpha$ ?

**Solution.** **Type I ( $\alpha$ ):** rejecting a true  $H_0$  — a false positive. **Type II ( $\beta$ ):** failing to reject a false  $H_0$  — a false negative (missing a real effect). Power =  $1 - \beta$ . With  $H_0$  = “no cancer”, a **Type II error** (missing an actual cancer) is far more serious than

a false alarm, which a follow-up test can resolve. So we deliberately raise  $\alpha$  (accept more false positives) to reduce  $\beta$  — i.e. we prefer a high-sensitivity screen. The two errors trade off at a fixed  $n$ ; increasing  $n$  reduces both.

**Q5. [2 marks]** Explain the difference between MLE and MAP estimation. Under what condition do they coincide?

**Solution.** **MLE** maximises the likelihood:  $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} P(D | \theta)$  — the parameter making the observed data most probable. It uses data only.

**MAP** maximises the posterior:  $\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} P(D | \theta)P(\theta)$  — it weights the likelihood by a prior.

They **coincide when the prior is uniform** (flat) over the parameter range, since then  $P(\theta)$  is a constant that does not affect the arg max. With large  $n$  the likelihood dominates any fixed prior, so they also converge asymptotically.

## Part B — Probability and Distributions (20 marks)

**Q6. [5 marks]** An IT department buys laptops from three vendors: 50% from A, 30% from B, 20% from C. The failure rates within warranty are 2%, 4% and 5% respectively. (a) What is the probability a randomly chosen laptop fails within warranty? (b) Given that a laptop has failed, what is the probability it came from vendor C? (c) Which vendor is the most likely source of a failed laptop?

**Solution.** (a) **Law of total probability.**

$$P(F) = 0.50(0.02) + 0.30(0.04) + 0.20(0.05) = 0.010 + 0.012 + 0.010 = \mathbf{0.032} \text{ (3.2\%).}$$

(b) **Bayes' theorem.**

$$P(C | F) = \frac{P(F | C)P(C)}{P(F)} = \frac{0.010}{0.032} = \mathbf{0.3125}.$$

(c)  $P(A | F) = 0.010/0.032 = 0.3125$ ;  $P(B | F) = 0.012/0.032 = \mathbf{0.375}$ ;  $P(C | F) = 0.3125$ . (They sum to 1 ✓.) **Vendor B** is the most likely source of a failed laptop, even though vendor C has the highest failure *rate* — B supplies more units. This is the base-rate effect.

**Q7. [5 marks]** A web server receives requests at an average rate of 4 per second. (a) Name the appropriate distribution and state its mean and variance. (b) Find  $P(\text{exactly 2 requests in a second})$ . (c) Find  $P(\text{at least 2 requests in a second})$ . (d) What is the probability of no requests in a 0.5-second window? Use  $e^{-4} = 0.0183$ ,  $e^{-2} = 0.1353$ .

**Solution.** (a) **Poisson**( $\lambda = 4$ ) — counts of independent rare events in a fixed interval. Mean = variance =  $\lambda = 4$ .

(b)  $P(X = 2) = \frac{e^{-4}4^2}{2!} = \frac{0.0183 \times 16}{2} = \mathbf{0.1465}.$

(c)  $P(X \geq 2) = 1 - P(0) - P(1) = 1 - e^{-4} - 4e^{-4} = 1 - 5(0.0183) = 1 - 0.0915 = \mathbf{0.9085}.$

(d) In half a second the rate is  $\lambda' = 4 \times 0.5 = 2$ .  $P(X = 0) = e^{-2} = \mathbf{0.1353}.$

**Q8. [5 marks]** API response times are normally distributed with  $\mu = 250$  ms and  $\sigma = 40$  ms. (a) What fraction of requests exceed 300 ms? (b) What fraction fall between 210 ms and 290 ms? (c)

The team wants to publish a “p90” latency figure (the value exceeded by only 10% of requests). Compute it. (d) If the SLA is “95% of requests under 320 ms”, is it met?

**Solution.** (a)  $z = (300 - 250)/40 = 1.25$ . Interpolating is not needed if we use  $z = 1.28 \rightarrow \Phi = 0.90$ ; using the supplied table at the nearest value  $z = 1.28$  gives  $P(X > 300) \approx 1 - 0.90 = \mathbf{0.10}$  (about 10%). (With a full table,  $\Phi(1.25) = 0.8944$ , giving 10.6%.)  
 (b)  $z_1 = (210 - 250)/40 = -1$ ,  $z_2 = (290 - 250)/40 = 1$ .  $P(-1 < Z < 1) = 0.8413 - (1 - 0.8413) = \mathbf{0.6826}$  (the 68% empirical rule).  
 (c) p90:  $\Phi(z) = 0.90 \Rightarrow z = 1.28$ .  $x = 250 + 1.28(40) = \mathbf{301.2}$  ms.  
 (d) The 95th percentile is  $z = 1.645$ :  $x = 250 + 1.645(40) = 315.8$  ms. Since  $315.8 < 320$ , **the SLA is met** — in fact 95% of requests complete within 315.8 ms, comfortably inside the 320 ms budget.

**Q9. [5 marks]** A new build process is claimed to have mean compile time 10 minutes. A sample of  $n = 16$  builds gives  $\bar{x} = 11.2$  min with  $s = 2.4$  min. (a) State  $H_0$  and  $H_1$  for a two-sided test. (b) Compute the test statistic and state the degrees of freedom. (c) Test at  $\alpha = 0.05$  using  $t_{0.025,15} = 2.131$ . (d) Construct the 95% confidence interval and confirm it agrees with your decision.

**Solution.** (a)  $H_0 : \mu = 10$  vs  $H_1 : \mu \neq 10$ .  
 (b)  $\sigma$  unknown and  $n$  small, so use the  $t$ -statistic with  $n - 1 = \mathbf{15}$  degrees of freedom:  

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{11.2 - 10}{2.4/4} = \frac{1.2}{0.6} = \mathbf{2.0}.$$
  
 (c)  $|t| = 2.0 < 2.131 = t_{0.025,15}$ , so we **fail to reject**  $H_0$  at the 5% level. The evidence is not strong enough to conclude the mean compile time differs from 10 minutes. (Note: “fail to reject” is not “prove  $H_0$  true”.)  
 (d) 95% CI:  $11.2 \pm 2.131 \times 0.6 = 11.2 \pm 1.279 = [\mathbf{9.92}, \mathbf{12.48}]$ . This interval *contains* 10, which is exactly equivalent to failing to reject  $H_0$  at  $\alpha = 0.05$ . ✓

## Part C — Bayesian Inference and Regression (20 marks)

**Q10. [6 marks]** A spam filter is being evaluated. Of 12 emails tested from a new sender, 9 were spam. (a) Give the MLE of the spam probability  $\theta$ . (b) Using a Beta(2, 2) prior, derive the posterior distribution and state its mean and MAP estimate. (c) Repeat with a Beta(1, 1) prior and comment on the relationship to the MLE. (d) Explain in one sentence what would happen to the influence of the prior if 1200 emails were observed with the same 75% spam rate.

**Solution.** (a) MLE =  $k/n = 9/12 = \mathbf{0.75}$ .  
 (b) Beta is conjugate to the binomial: posterior = Beta( $\alpha + k$ ,  $\beta + n - k$ ) = Beta(2 + 9, 2 + 3) = Beta(**11, 5**).

$$\text{Posterior mean} = \frac{\alpha'}{\alpha' + \beta'} = \frac{11}{16} = \mathbf{0.6875}; \quad \text{MAP} = \frac{\alpha' - 1}{\alpha' + \beta' - 2} = \frac{10}{14} = \mathbf{0.7143}.$$

Both are pulled below the MLE of 0.75, toward the prior mean of 0.5 (*shrinkage*).

(c) With Beta(1, 1) (the uniform prior): posterior = Beta(10, 4); mean =  $10/14 = 0.714$ ; MAP =  $9/12 = \mathbf{0.75}$  = the MLE. A flat prior makes MAP and MLE identical.

(d) With  $n = 1200$  and  $k = 900$ , the posterior would be Beta(902, 302) with mean 0.7492 — the two pseudo-counts from the prior are swamped by 1200 real observations, so the prior’s influence becomes negligible and the Bayesian and frequentist answers converge.

**Q11. [8 marks]** The table gives hours of CPU profiling effort  $x$  and the resulting percentage speed-up  $y$  for six modules:

|     |   |   |    |    |    |    |
|-----|---|---|----|----|----|----|
| $x$ | 2 | 4 | 6  | 8  | 10 | 12 |
| $y$ | 5 | 9 | 14 | 17 | 22 | 27 |

(a) Compute  $\bar{x}$ ,  $\bar{y}$ ,  $S_{xx}$ ,  $S_{yy}$  and  $S_{xy}$ . (b) Find the least-squares regression line  $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ . (c) Compute the correlation coefficient  $r$  and the coefficient of determination  $R^2$ ; interpret  $R^2$  in one sentence. (d) Predict the speed-up for 9 hours of effort. (e) State one reason why predicting at  $x = 40$  would be unwise.

**Solution.** (a)  $\bar{x} = \frac{2+4+6+8+10+12}{6} = 7$ ;  $\bar{y} = \frac{5+9+14+17+22+27}{6} = \frac{94}{6} = 15.6667$ .

| $x$ | $y$ | $x - \bar{x}$                                     | $y - \bar{y}$ | $(x - \bar{x})^2$ | $(y - \bar{y})^2$ | $(x - \bar{x})(y - \bar{y})$ |
|-----|-----|---------------------------------------------------|---------------|-------------------|-------------------|------------------------------|
| 2   | 5   | -5                                                | -10.6667      | 25                | 113.778           | 53.333                       |
| 4   | 9   | -3                                                | -6.6667       | 9                 | 44.444            | 20.000                       |
| 6   | 14  | -1                                                | -1.6667       | 1                 | 2.778             | 1.667                        |
| 8   | 17  | 1                                                 | 1.3333        | 1                 | 1.778             | 1.333                        |
| 10  | 22  | 3                                                 | 6.3333        | 9                 | 40.111            | 19.000                       |
| 12  | 27  | 5                                                 | 11.3333       | 25                | 128.444           | 56.667                       |
|     |     | $S_{xx} = 70$ $S_{yy} = 331.333$ $S_{xy} = 152.0$ |               |                   |                   |                              |

(b)  $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{152}{70} = 2.1714$ ;  $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 15.6667 - 2.1714(7) = 15.6667 - 15.2000 = 0.4667$ .

$$\hat{y} = 0.4667 + 2.1714x$$

(c)  $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{152}{\sqrt{70 \times 331.333}} = \frac{152}{\sqrt{23193.3}} = \frac{152}{152.294} = 0.9981$ .  $R^2 = r^2 = 0.9961$ .

*Interpretation:* 99.6% of the variation in speed-up across these modules is explained by the linear relationship with profiling hours — an almost perfectly linear fit.

(d)  $\hat{y}(9) = 0.4667 + 2.1714(9) = 0.4667 + 19.543 = 20.01\%$ .

(e)  $x = 40$  is far outside the observed range  $[2, 12]$ : this is **extrapolation**. The linear model predicts 87.3% speed-up, but nothing in the data supports linearity that far out, and diminishing returns almost certainly set in (there is also a physical ceiling — Amdahl's law bounds the achievable speed-up regardless of effort).

**Q12. [6 marks]** Using the regression from the previous question: (a) Compute the fitted values and residuals for  $x = 2$ ,  $x = 6$  and  $x = 12$ . (b) Compute SSE and SST, and verify  $R^2 = 1 - \text{SSE}/\text{SST}$ . (c) Test whether the slope is significantly different from zero at  $\alpha = 0.05$ , given  $t_{0.025,4} = 2.776$  and

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\text{SSE}/(n-2)}{S_{xx}}}.$$

**Solution.** (a)  $\hat{y} = 0.4667 + 2.1714x$ :

| $x$              | $y$ | $\hat{y}$                  | residual $e = y - \hat{y}$ |
|------------------|-----|----------------------------|----------------------------|
| 2                | 5   | $0.4667 + 4.3429 = 4.8095$ | +0.1905                    |
| 4                | 9   | 9.1524                     | -0.1524                    |
| 6                | 14  | 13.4952                    | +0.5048                    |
| 8                | 17  | 17.8381                    | -0.8381                    |
| 10               | 22  | 22.1810                    | -0.1810                    |
| 12               | 27  | 26.5238                    | +0.4762                    |
| $\sum e \approx$ |     |                            | 0.0000 ✓                   |

(b)  $\text{SSE} = 0.0363 + 0.0232 + 0.2548 + 0.7024 + 0.0328 + 0.2268 = \mathbf{1.2763}$ .  $\text{SST} = S_{yy} = \mathbf{331.333}$ .

$$1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{1.2763}{331.333} = 1 - 0.00385 = \mathbf{0.9961} = R^2. \checkmark$$

(c)  $\text{SE}(\hat{\beta}_1) = \sqrt{\frac{1.2763/4}{70}} = \sqrt{\frac{0.3191}{70}} = \sqrt{0.004558} = 0.0675.$

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} = \frac{2.1714}{0.0675} = \mathbf{32.2}.$$

Since  $32.2 \gg 2.776$ , we **reject**  $H_0 : \beta_1 = 0$ : the slope is highly significantly different from zero, and profiling effort is a genuine predictor of speed-up in this data.